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Roberto Fernández-Barrios
Faculty of Engineering, University of Deusto
Avda. de las Universidades 24, 48007 Bilbao, Spain
roberto.fernandez.b@deusto.es

Editors
npj Quantum Information

Re: Article submission to the Collection "Quantum machine learning: understanding capabilities, limitations, and perspectives for quantum advantage"

Dear Editors,

We submit "Conditional validity of quantum event classifiers under collider systematics and quantum estimation uncertainty" for consideration as an Article in the Collection named above. Rather than presupposing quantum advantage, the manuscript asks a prior question for deployable QML: under what observable information can a performance or scientific-inference claim be supported when both the scientific environment and the quantum estimator are uncertain?

The work makes a connection that is not captured by nominal QML benchmarking. Collider systematics determine whether the scientific claim is identifiable; finite-shot and noisy quantum-kernel evaluation additionally randomizes the implemented Gram matrix, refit, calibration, and operating point. We integrate these layers in one fail-closed framework and distinguish claims about the realized deployment from claims anchored to an ideal quantum kernel. This allows the paper to say exactly what quantum measurement changes without attributing ordinary stochasticity exclusively to quantum systems.

We have updated the positioning against the closest new literature. Brown, Spannowsky and Williams (arXiv:2608.11330) now evaluate collider QML models under controlled detector-inspired feature smearing. Our claim is therefore not priority for studying collider shift. The distinct contribution is the combination of official physically parameterized nuisances, including rate-only effects; an explicit I0–I3 observable-information hierarchy; fail-closed anytime-valid certification; propagation to signal-strength inference; and finite-shot/noisy Gram estimation treated as part of the realized deployment.

The results directly address the Collection's emphasis on capabilities and limitations. Matched classical controls remove every apparent quantum-specific performance and sensing effect, so we claim no quantum advantage. What remains QML-specific is a measurement-induced deployment uncertainty whose effects propagate from the realized Gram matrix through refitting, calibration, thresholding and claim resolution. The eight-qubit experiment is classically simulable, and the QPU result is only a micro-scale integration/fail-closed consistency demonstration. The collider setting is a stringent scientific deployment environment: it supplies physics weights, invisible yield shifts, control regions, finite templates, and downstream signal-strength inference. Evidence spans disjoint simulated worlds, CMS open data, and a deliberately micro-scale IBM QPU demonstration, with all scope limits stated explicitly.

The manuscript combines formal identifiability and anytime-valid per-claim guarantees for fixed I2 label streams, conditional on the frozen finite audit population under the declared with-replacement protocol, with separately coverage-gated I3 inference and calibrated empirical ledgers, registered falsifiers, immutable manifests, public code and data artifacts, and a complete correction trail. Several registered falsifiers fired; each adverse result was retained and used to narrow the claims. We believe this combination of rigorous negative controls, quantum deployment semantics, and physics-level validation will be useful to the journal's interdisciplinary QML readership.

For transparency, the same author group has released distinct public material, "Sharp Target-Domain Certificates for Quantum-Kernel Advantage under Distribution Shift" (Zenodo DOI 10.5281/zenodo.21776862). It is currently available as public preprint/Zenodo material and has not been submitted to any journal. That paper studies partial identification and certification of predictive advantage under target-domain shift. The present paper instead studies scientific claim validity under collider systematics, incomplete information, downstream physics inference and measurement-induced quantum deployment uncertainty. The works share no datasets, experiments, theorems, estimands, codebase, or reported results, and are not manuscripts simultaneously under consideration.

This manuscript is not under consideration elsewhere. All authors have approved the manuscript and its submission. The authors declare no financial or non-financial competing interests.

Sincerely,

Roberto Fernández-Barrios
Corresponding author, on behalf of all authors